

Ex.No:3 Assessment of IT Project Risk Analysis using open-source tools.

AIM:

To assess and analyze IT project risks using open-source risk analysis tools, identify potential risks, evaluate their likelihood and impact, and develop appropriate mitigation strategies to improve project success and reduce risk exposure.

ALGORITHM:

- Start the Mendix Studio Pro and create a new blank project named **RiskManagementSystem**.
- Create a **RiskManagement** module and open its Domain Model.
- Create a **Risk** entity to store project risk information.
- Add attributes such as **RiskName, Description, Probability, Impact, RiskScore, MitigationPlan, and RiskOwner**.
- Create **RiskCategory** and **RiskStatus** enumerations and add them to the Risk entity.
- Generate the **Risk Overview** and **Risk New/Edit** pages automatically.
- Configure the application navigation to provide access to the **Risk Overview** page.
- Create risk records and calculate the risk score using **Risk Score = Probability × Impact**.
- Test the **Create, Read, Update, and Delete (CRUD)** operations on the risk records.
- Verify the risk information and analyze the risks based on their **probability, impact, score, category, and status**, then stop.

PROCEDURE:

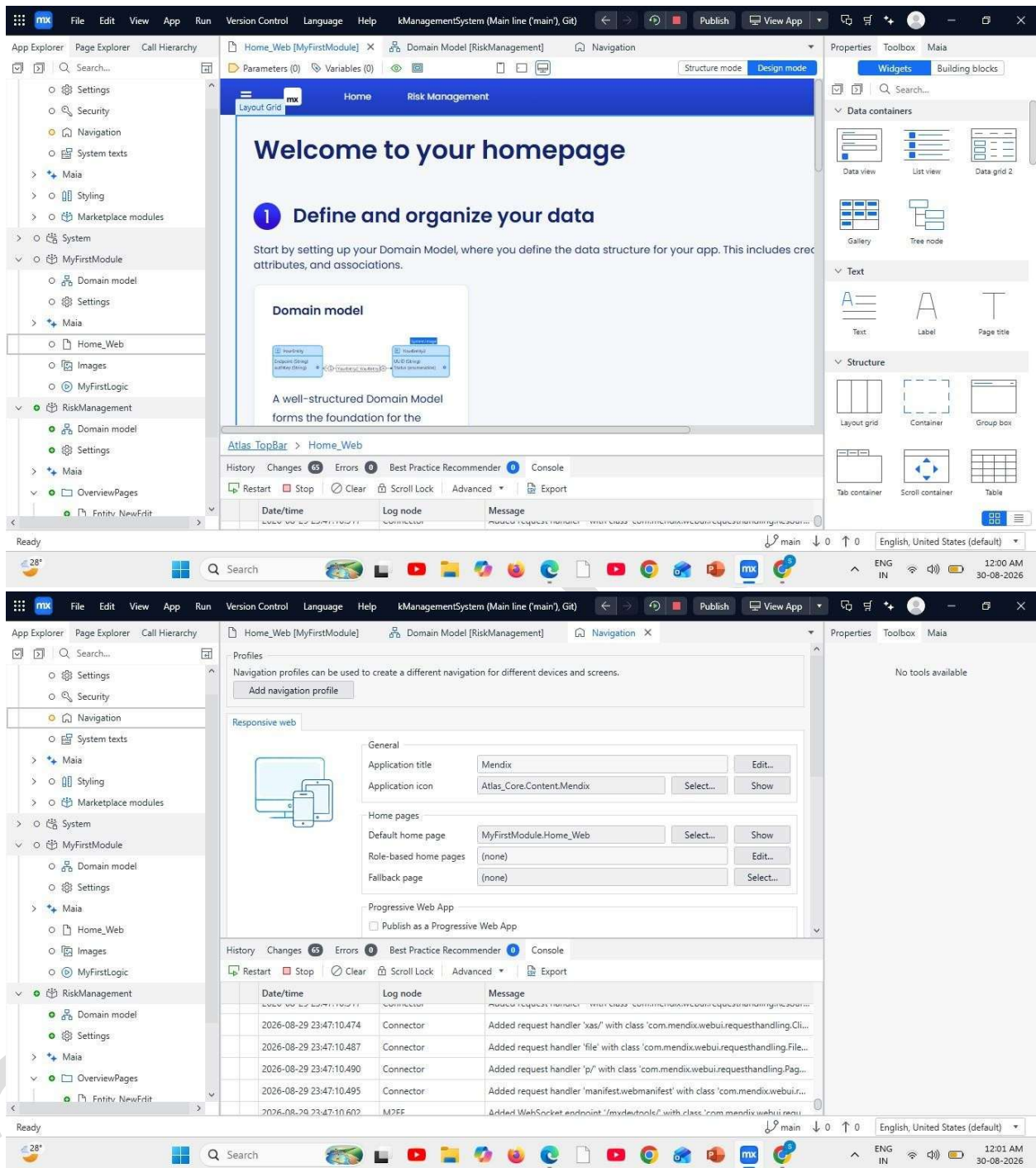
- Open Mendix Studio Pro, create a Blank App, and name it **RiskManagementSystem**.
- Add a module named **RiskManagement** and open the Domain Model.
- Create the Risk entity and add the required risk-related attributes.
- Create the **RiskCategory** enumeration with Technical, Schedule, Cost, Resource, Security, and Quality values.
- Create the **RiskStatus** enumeration with Open, UnderReview, Mitigated, and Closed values, and add both enumerations to the Risk entity.
- Generate the Risk Overview and Risk New/Edit pages from the Risk entity.

- Configure System → Navigation and add Risk Management to open the Risk Overview page.
- Run the application and create sample risk records with probability, impact, risk score, category, status, owner, and mitigation plan.
- Verify the records and test Create, Read, Update, and Delete operations.
- Confirm that all risk details and risk scores are displayed correctly and that the Risk Management System is functioning successfully.

OUTPUT:

The screenshot displays the Mendix IDE interface and the running application. The IDE window shows the 'Home_Web' page with a 'Welcome to your homepage' message and a 'Domain model' section. The console shows 0 errors, 1 deprecation, and 17 warnings. The application running in the browser shows the 'Entity' overview page with a table of risk records.

Risk name	Description	Probability	Impact	Risk score	Mitigation plan	Risk owner
Server Failure	Database server ma...	5	4	20	Daily Backup	Admin Team
Data Breach	Unauthorized acces...	5	5	25	Encryption and acce...	Security team
Network Failure	Network connectiv...	3	4	12	Backup network con...	IT Team
Budget overrun	Project expenses m...	4	3	12	Regular budget mo...	Finance
Hardware damage	Project hardware m...	2	5	10	Maintain backup ha...	Infrastructure



RESULT:

The IT Project Risk Management System is successfully developed using Mendix, and the identified project risks can be recorded, assessed, monitored, and managed based on their probability, impact, risk score, and mitigation status.